

Illustrative scheduler decision — catalog manifests

$$S(m) = F_{cap} - \alpha M_{cost} - \beta L_{load} - \gamma E_{cost} - \delta P_{evict} + \eta F_{fut}$$

$[\alpha = 0.20, \beta = 0.25, \gamma = 0.10, \delta = 0.30, \eta = 0.35]$

Stage: math derivation | B_{RAM} = 8.0 GB | Resident: research-expert

Candidate	F_{cap}	αM	βL	γE	δP	ηF	Total	Status
math-expert	+0.96	-0.06	-0.05	-0.02	0.00	0.00	+0.83	SELECTED
coding-expert	+0.05	-0.08	-0.06	-0.02	0.00	+0.35	+0.25	AVAILABLE
research-expert	+0.05	-0.08	0.00	-0.02	0.00	0.00	-0.05	RESIDENT
general-reasoner	+0.46	-0.18	-0.11	-0.03	-0.08	0.00	+0.06	OUTSCORED

general-reasoner admissible under 8.0 GB budget, but outscored by memory cost (7.1 GB) and eviction deficit (2.2 GB deficit vs 4.9 GB free RAM).